

Data Abstraction & 3-Level Schema Architecture

Q. What is Data Abstraction in a DBMS? Explain the three levels of data abstraction with a suitable example.

> **Definition to Remember**

1. The Three Levels of Data Abstraction

VIEW LEVEL (External) End Users see their specific views

LOGICAL LEVEL (Conceptual) Designers see the full structure

PHYSICAL LEVEL (Internal) DBA sees raw storage details

1. Physical Level (Internal Level) *Lowest Level*

2. Logical Level (Conceptual Level) *Middle Level*

3. View Level (External Level) *Highest Level*

2. Example University Database

| Level | What is Seen |

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

> `Phy